

Alyssa Dang

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Expected May 2028

B.S. in Human-Computer Interaction + Cognitive Science (Machine Learning) | GPA: 4.0/4.0

- Relevant Coursework: Computational Neuroscience, Designing Human-Centered Software (in progress), Principles of Imperative Computation (in progress), Cognitive Psychology

RESEARCH & TECHNICAL EXPERIENCE

Research Assistant, Laboratory of Computational Perception

Aug 2026 - Present

- Collaborate with a small research team under faculty supervision to train deep neural networks in PyTorch, studying how task objectives shape the models' learned representations of body position and movement

V1 Neural Decoding & Systems Identification, Computational Neuroscience Lab

Apr 2026

- Analyzed mouse primary visual cortex (V1) Neuropixel recordings (Allen Institute/DANDI) across drifting-grating and Gabor stimuli using Gaussian-kernel smoothing, tuning curves, spatial receptive field mapping, and Linear-Nonlinear-Poisson (LNP) model validation via spike-triggered averaging
- Applied Principal Component Analysis (PCA) via SVD on zero-centered population response matrices, identifying a low-dimensional manifold where 6 principal components captured 95% of variance
- Developed OLS linear regression decoders with circular target encodings (80/20 train/test split), demonstrating significantly higher decoding accuracy for stimulus orientation over direction (31° vs. 67° mean test error)

LTP/LTD Circuit Simulation, Computational Neuroscience Lab

Feb 2026

- Built a Leaky Integrate-and-Fire (LIF) circuit model in Python (NumPy, Matplotlib, Jupyter Notebook) simulating AMPA/NMDA receptor kinetics, calcium current, and membrane voltage dynamics of an excitatory synapse
- Implemented Spike-Timing-Dependent Plasticity (STDP) via calcium-threshold learning rules modulating synaptic conductance (LTP/LTD) within a biologically accurate 20ms spike-timing window
- Validated model behavior against published literature through comparative plots of I-V curves and conductance history

PROJECTS

NBA Outcome Prediction & Optimization Model, ML & Probabilistic Modeling

Apr 2026

- Modeled NBA home win probabilities in R using Random Forest classifiers across multiple architectures (all predictor variables, top predictors, and home team predictors)
- Evaluated models via 5-fold cross-validation, achieving a 0.102 Brier score on top feature sets, and quantified prediction uncertainty via tree standard deviations to guide expected-value decision-making

ADDITIONAL EXPERIENCE

Product & UI/UX Designer, CravingChef (iOS App Store)

Jul 2026 – Aug 2026

- Partnered directly with the founder to shape product decisions as UI/UX designer, synthesizing user feedback and behavioral analytics into interface decisions and establishing human-centered research workflows applicable to neurotechnology and assistive human-machine interfaces

Project Lead, Honey Haven — CMU Game Creation Society

Jan 2026 – Apr 2026

- Directed a small cross-functional team (design, narrative, software) to pitch, scope, and ship a project in one semester; won Game Creation Society Gold Award

SKILLS

Technical: Python, PyTorch, NumPy, Matplotlib, R, C, Dimensionality Reduction (PCA, SVD), Linear Regression Decoders, LNP Modeling, LIF Neurons, Spike-Triggered Averaging, Jupyter, HTML/CSS, JavaScript, Git Command Line

Research & Signal Processing: Neural Data Processing (Neuropixels/DANDI), Signal Filtering, Receptive Field Mapping, Computational Modeling, Cross-Validation, Usability Testing

Collaboration & Product: Figma, Cross-Functional Collaboration, Stakeholder Communication, Google Workspace, Adobe Creative Suite